

Instructions

- Answer all eight questions. Show sufficient working for method marks.
- A calculator may be used unless a question states otherwise.
- Write clearly in the spaces provided. Additional pages may be attached.

1

Convert the binary number to denary and explain the place-value method used.

[4 marks]

1011 0110₂

2

Describe how a hash table resolves collisions, naming one strategy and its drawback.

[6 marks]

3

Write a function that returns the maximum value in a list without using a built-in max.

[5 marks]

4

Explain why binary search requires a sorted list, and state its complexity.

[5 marks]

$$O(\log n)$$

5

Compare compiled and interpreted execution, giving one advantage of each.

[5 marks]

6

Describe the purpose of a primary key in a relational database and why duplicates break it.

[4 marks]

7

Explain recursion using a worked example, and state the role of the base case.

[6 marks]

8

Identify the logic error in the described loop and explain the correction.

[5 marks]
